

Kanu Madhok

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PROFESSIONAL EXPERIENCE

Walmart Data Ventures

Chicago, IL · Sep 2024 – Present

Senior Data Analyst

- Designed and shipped to production a self-service analytics agent over 67 BigQuery tables with sub-agent orchestration and human-in-the-loop validation (Context Researcher → SQL Drafter → Validator → Devil's Advocate); 8 deterministic SQL rules and a 23-case golden-query evaluation suite. Partnered with Product and Data Science to translate ambiguous business questions into well-scoped, agent-ready specs. Approved by Sr. Director and product leadership and the only AI skill in active use by business teams on Data Ventures.
- Built and shipped to production an autonomous Jira ticket-resolution agent applying agentic LLM patterns (tool use, multi-step reasoning, orchestration): 6-gate triage, MCP-driven context gathering, agentic RAG with FAISS + BM25 retrieval over 11,000 historical SQL queries, and self-healing execution with a 10-retry budget. Closed 400+ requests across 8 categories over 15 months — cut routine turnaround from 30–60 minutes to under 10 minutes — automating high-complexity work that previously required human judgment.
- Built a registry-governed semantic layer over BigQuery where a single LLM call maps the question to strict JSON intent (structured-output generation; the model never writes SQL); a deterministic gate validates and canonicalizes intent before a deterministic compiler emits the SQL — same question, same answer, every time, with the one probabilistic step regression-testable. Made sound custom-vs-off-the-shelf calls (retired a parallel CubeJS comparison flow that was double compute for offline value).
- Built and deployed to production a headless KPI monitor on a 6-hour cron that detects threshold breaches, runs root-cause analysis, and posts Slack alerts; end-to-end observability via decision logging and BigQuery state persistence. Designed a v2 as a 3-tier event-driven platform self-configuring via a BigQuery event registry — new event types ship without code deploys.
- Replaced ~20 recurring recruitment workflows with end-to-end automation on a 68-table BigQuery platform; stratified-sampling pipelines recruited 280,000+ panelists across 29 categories with zero duplicates, turning a recurring 3-hour task into ~3 minutes of setup.

University of Chicago — Data Science Institute

Chicago, IL · Jan – Sep 2024

Graduate Researcher, Data & Democracy Initiative

- Persona-driven multi-LLM (OpenAI + Gemini) chatbot for a 500+ participant donation experiment; controlled A/B tests persuaded 15% to donate to an opposing cause.

FTI Consulting — UChicago MSADS Capstone

Chicago, IL · Jan – Aug 2024

- RAG-to-proposal generator deployed on AWS EC2; cut proposal time ~30%; Best in Show — UChicago MSADS Capstone.

Innovare (EdTech Startup)

Chicago, IL · Oct 2022 – Mar 2023

Data Scientist

- Logistic-regression risk model + Looker dashboards for 10+ districts (5,000+ students); retention +7%, failures –15%.

SELECTED PROJECT

- Live deployed text-to-SQL copilot (Streamlit + Gemini 2.5 + LangChain-orchestrated RAG over historical queries, BigQuery execution, auto-visualization): live demo — <https://sql-rag-frontend-simple-481433773942.us-central1.run.app/>

SKILLS

Agent & evaluation stack: agentic LLM patterns (tool/function calling, multi-step reasoning, orchestration), MCP (Model Context Protocol) server design, sub-agent orchestration, context engineering, prompt engineering, RAG (FAISS + BM25 + RRF), structured-output generation, evaluation suites (golden sets, human-in-the-loop validation), multi-model routing, event-driven architectures, decision logging / observability

LLMs: Claude (Opus / Sonnet / Haiku via Wibey), Gemini (2.5 Pro / Flash via Vertex AI), OpenAI

APIs / triggers / glue: FastAPI (REST), Cloud Run, BigQuery scheduled queries, Slack webhooks, cron, Playwright SSO, JSONL state stores, CI/CD (pytest, version control)

Data: BigQuery, dbt, SQL, Python, PySpark, Airflow, Databricks, MySQL

EDUCATION

The University of Chicago — M.S., Applied Data Science (GPA 4.0)

Chicago, IL

Loyola University Chicago — B.S., Information Systems & Economics (GPA 3.5)

Chicago, IL